Data Analysis Report

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Report

Final report for eye tracking course.

Source code and data: https://github.com/jnagasawa/ET-report

Chapter 1

Background

A robust leftward bias in early visual exploration has been documented across diverse tasks, including free scene viewing, line bisection, face perception, and two-alternative forced choice paradigms (Foulsham et al. 2013; Ossandón et al. 2014). In speakers of left-to-right (LTR) languages, the proportion of initial fixations landing on the left side of a display typically falls between 60% and 93%, depending on the task and stimulus material (Ehinger et al. 2026). Because this preference operates independently of image content, it can substantially reduce the sensitivity of paradigms designed to measure content-driven attentional biases.

The origin of the leftward bias remains debated. A dominant account links it to the right-hemisphere lateralization of the attentional network, which preferentially represents the left hemifield (Corbetta and Shulman 2002). This explanation does not, however, readily account for cross-linguistic variation: speakers of Arabic and Hebrew show a weaker or reversed preference in line bisection and related tasks (Chokron and Imbert 1993; Rashidi-Ranjbar et al. 2014), suggesting that habitual reading direction modulates the baseline spatial asymmetry.

Afsari, Ossandón, and König (2016) examined this modulation directly using a priming paradigm. Bilingual participants read either RTL or LTR text passages before freely viewing photographs. Native RTL speakers (Arabic, Urdu, or Persian) showed an early rightward shift in fixation after reading RTL primes, but a leftward shift after LTR primes. Native LTR speakers who had acquired an RTL language later in life did not show comparable flexibility. These results indicate that reading direction influences spatial scanning dynamically, and that the effect depends on habit formation during native language acquisition rather than current language use alone.

Complementary evidence concerns the consequences of spatial biases for experimental design. Ehinger et al. (2026) measured first-fixation preferences in a concurrent two-stimulus paradigm with stimulus pairs placed at six angular positions relative to a central fixation cross. In the standard horizontal (9 o'clock / 3 o'clock) arrangement, German-speaking participants directed their first saccade to the left stimulus on approximately 80% of trials, with a comparable upward bias in the vertical arrangement. A diagonal placement at 1 o'clock and 7 o'clock yielded no net spatial preference, because the leftward and upward biases cancelled. Simulations based on a Bradley-Terry model showed that eliminating the spatial bias can increase statistical power by 30% or more for small attentional effects.

1.1 Motivation

The literature on spatial scanning biases has been developed almost exclusively with LTR-reading populations, and spatial biases are routinely counterbalanced rather than measured as a variable of interest. As a result, little is known about the baseline first-fixation behavior of native RTL speakers in concurrent two-stimulus paradigms. Afsari et al. (2016) showed that reading direction primes can shift the spatial bias in bilingual RTL speakers, but the baseline behavior in the absence of explicit language priming has received less attention.

The findings of Ehinger et al. (2026) raise a further question: if RTL speakers have a reduced or reversed leftward bias, their data from a standard horizontal arrangement may already be relatively unbiased, whereas LTR speakers' data from the same arrangement would suffer substantial power loss. Characterizing the bias across reading-direction groups would therefore inform stimulus placement decisions in studies that test mixed-language samples.

1.2 Research Questions and Hypotheses

We ask whether native RTL speakers (Arabic and Persian) differ from native LTR speakers in their first-fixation behavior when viewing pairs of visual stimuli in a horizontal arrangement.

We expect LTR speakers to replicate the well-documented leftward preference. For RTL speakers, we predict a reduced leftward bias, consistent with line bisection and chimeric face studies. Whether the bias reverses entirely or merely diminishes is an open empirical question.

Dependent measures: first fixation proportion (the proportion of trials on which the participant fixated each side first) and first fixation reaction time (latency from stimulus onset to the first fixation on either stimulus).

Chapter 2

Experimental Methods

2.0.1 Participants

Data collection is ongoing. The current dataset includes 14 participants: 9 who selected a left to right (LTR) language (German or English) at the start of the session, and 5 who selected a right to left (RTL) language (Arabic or Persian). Group membership follows this self selected session language rather than a separate native language screening; participants were recruited as native speakers of the language they chose. Age, sex, and other demographic information will be added once collection is complete.

2.0.2 Materials and stimuli

Stimuli were 16 images (000.png through 015.png) organized into eight fixed pairs. Each pair was shown six times across the session, and its position in this cycle ran independently of the six trial block structure described below, so a given pair was not tied to any particular story segment.

On each trial, the two images of a pair appeared at the same time on opposite sides of a central fixation point, along one of six axis orientations spaced 30° apart (0°, 30°, 60°, 90°, 120°, 150° from horizontal). This follows the arrangement used by Ehinger et al. (2026). Within a block of six trials, each of the six orientations occurred exactly once, in an order randomized per participant.

To keep participants engaged with both images rather than only the one they looked at first, they were asked to count the circular shapes in the pair, distinguished from a similarly sized decagon shown once during the instructions. After each trial the experimenter entered the spoken count (0 to 3) by key press.

Alongside the eye tracking task, participants read a children's story (The Ugly Duckling) presented in nine segments. The first eight were shown one at a time, each before one six trial block; the ninth (the story's resolution) was shown once after the final trial, followed by five comprehension questions asked aloud by the experimenter and recorded manually. The story gave participants sustained exposure to text in their session language and a natural framing for the task; the comprehension questions served as an attention check rather than a scored outcome.

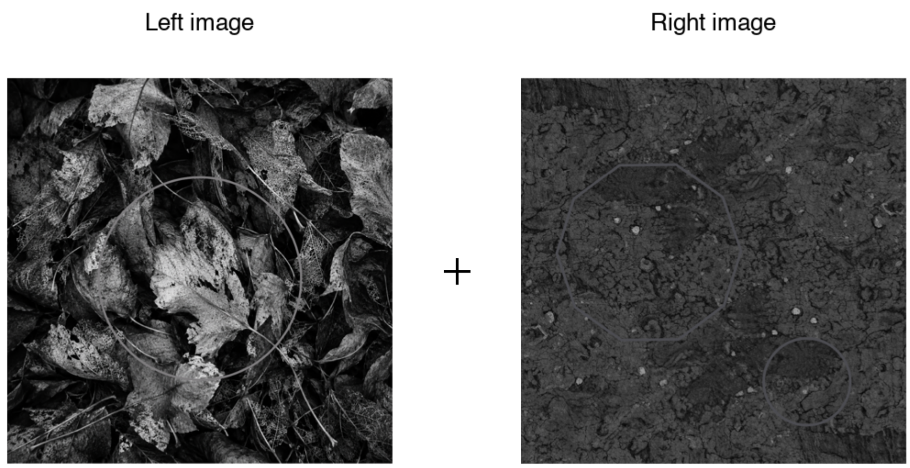

Figure 2.1: Example stimulus pair (pair 1 of 8) as presented on screen. Faint outline shapes embedded in the images are the circles participants were asked to count; the central cross marks the fixation point, not the on-screen fixation dot itself.

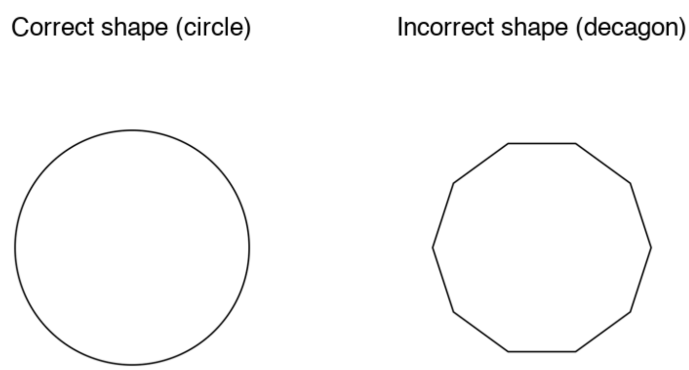

Figure 2.2: Shape distinction shown once during the instructions: the correct circle to count against the similarly sized incorrect decagon.

2.0.3 Study design

The experiment used a between subjects design with reading direction group (LTR vs RTL) as the main factor. Each participant completed 48 trials in 8 blocks of 6, with one story segment shown at the start of each block. Within a block, the six trials sampled the six axis orientations in random order.

2.0.4 Experiment (implementation)

The task was built in OpenSesame 4.1.14 and run with a GazePoint GP3 HD eye tracker via PyGaze, sampling at 60 Hz. Viewing distance was fixed at 57 cm from a 1536 x 864 px display (33.8 x 27.1 cm). Before the main task, participants chose their session language from four options (Deutsch, English, Arabic, Farsi) with a number key; this choice set both the group assignment and the language of the story text and on screen instructions.

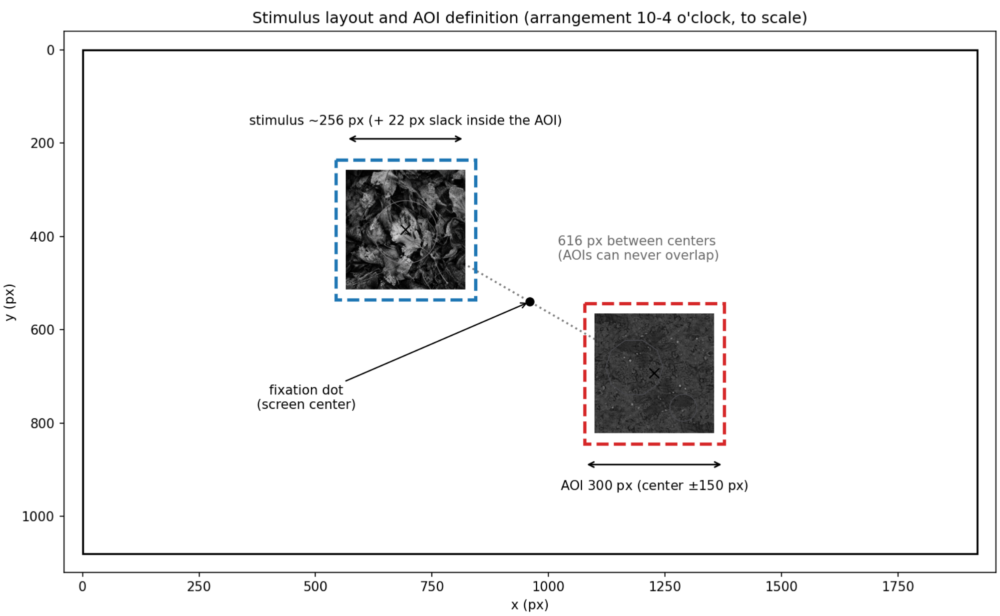

Figure 2.3: Stimulus layout and area-of-interest (AOI) definition for one arrangement (10-4 o'clock), drawn to scale. The two images are always equidistant from central fixation, along one of six axis orientations spaced 30° apart; a 150 px radius around each image center defines the AOI used for first-fixation detection.

Each trial opened with a central fixation check: the script polled the tracker for up to 1.5 s for a gaze sample within 100 px of screen center, then continued regardless of whether fixation was reached. A fixed 1000 ms delay followed before the image pair appeared. Once the pair was on screen, the tracker was polled for 5000 ms; a gaze sample within 150 px of either image's center was logged as the first fixation, recording its side, latency, and coordinates, but the pair stayed visible for the full 5 s regardless of when a fixation was detected. Calibration was performed once at the start of the session, with per participant accuracy and precision logged automatically.

2.1 Analysis methods

2.1.1 Preprocessing

The first fixation on each trial was identified from the raw gaze stream rather than taken from the value logged online during the session. For each trial, samples between stimulus onset and image offset were grouped into fixations by the tracker's own fixation index, and the first fixation starting at or after onset that fell inside a 150 px square area of interest around either stimulus was taken as the response, recording its side and latency. This square matches the margin the experiment script itself used for online detection. Two subjects (43 and 44) were recorded before a display scaling fix was applied to the script, which left their online-logged first fixation side unusable; the raw-stream recomputation is unaffected and was used for all subjects.

Trials were excluded when no fixation landed in either area of interest, or when the first fixation started less than 80 ms after onset, since a saccade that fast is planned before the stimulus appears rather than in response to it. No upper latency cutoff was applied. Slow first fixations were not random: they concentrated in arrangements without a stimulus in the upper left, so removing them would have removed a real part of the effect rather than noise.

Because the six stimulus arrangements are the six diameters of a clock face, a trial's outcome was recoded as the clock hour fixated first rather than a fixed left or right label. Left/right and up/down bias were then derived from this hour, each skipping the one arrangement lying on the axis perpendicular to it (12 to 6 for left/right, 9 to 3 for up/down), since those positions carry no information for that contrast.

2.1.2 Quality control

Before analysis, each recording was checked for complete trial markers (48 stimulus onsets, offsets, and fixation dot events), image display duration close to the intended 5 s, and effective sampling rate close to 60 Hz with few gaps longer than 50 ms. Calibration quality was estimated per trial from gaze recorded during the central fixation dot: accuracy as the mean distance from screen center, and precision as the RMS spread around each trial's own mean. These were compared across subjects with boxplots and an x/y offset scatter to catch systematic drift, and a small number of trials per subject were plotted as full scanpaths to confirm that gaze actually landed inside the areas of interest and that the arrangement positions matched the intended layout.

2.1.3 Analysis pipeline

Group comparisons used a hierarchical bootstrap: subjects were resampled first, then trials within each resampled subject, since trials from the same subject are not independent. The point estimate is the unweighted mean of subject means, with the 2.5th and 97.5th percentiles of the resampled distribution as a confidence interval. This was applied to left/right and up/down fixation proportions per group, and to the full 12 hour clock-face profile.

A logistic regression in the style of a Bradley-Terry model estimated the probability of fixating the clockwise-numbered stimulus first as a function of arrangement and reading direction group, giving a single test of whether the two groups differ once arrangement is accounted for. First-fixation latency was examined separately by arrangement and group, since the spatial bias also showed up as a difference in response speed rather than only in which side was chosen.

Chapter 3

Chapter 4

- 1-paragraph summary
- How do findings relate to literature?
- Limitations
- Challenges during the project
- Outlook / improvements / lessons learned

Chapter 5

Contribution table

Task	Junichi	Nikhil	Sanaz	Shehab
Background Literature	-	-	—	—
Experiment Design	-	-	—	—
Stimulus Design	—	—	-	-
Piloting	—	—	—	-
Data-Recording	—	-	-	—
Non-Final-Talk presenting (who talks)	—	-	—	-
Non-Final-Talk presenting (who prepares)	—	-	—	-
Final-Talk presenting (who talks)	—	—	—	-
Final-Talk presenting (who prepares)	—	-	-	-
Data Analysis Scripts	—	-	—	-
Report Writing	—	—	-	-
Report Lectorate	—	-	-	-

References

Chokron, Sylvie, and Michel Imbert. 1993. “Influence of Reading Habits on Line Bisection.” *Cognitive Brain Research* 1 (4): 219–22. https://doi.org/10.1016/0926-6410(93)90005-P.

Corbetta, Maurizio, and Gordon L. Shulman. 2002. “Control of Goal-Directed and Stimulus-Driven Attention in the Brain.” *Nature Reviews Neuroscience* 3 (3): 201–15. https://doi.org/10.1038/nrn755.

Ehinger, Benedikt, Maximilian Dominik Bernhardt, Simon Enkel, et al. 2026. *A Simple Spatial Bias Modification to Increase Statistical Power in Concurrent Two Alternatives Choice Tasks*.

Foulsham, Tom, Alasdair Gray, Eleni Nasiopoulos, and Alan Kingstone. 2013. “Leftward Biases in Picture Scanning and Line Bisection: A Gaze-Contingent Window Study.” *Vision Research* 78: 14–25. https://doi.org/10.1016/j.visres.2012.12.001.

Ossandón, José P., Selim Onat, and Peter König. 2014. “Spatial Biases in Viewing Behavior.” *Journal of Vision* 14 (2): 20. https://doi.org/10.1167/14.2.20.

Rashidi-Ranjbar, Neda, Mahdi Goudarzvand, Shahram Jahangiri, Peter Brugger, and Tobias Loetscher. 2014. “No Horizontal Numerical Mapping in a Culture with Mixed-Reading Habits.” *Frontiers in Human Neuroscience* 8: 72. https://doi.org/10.3389/fnhum.2014.00072.